

# Outlet water temperature prediction of energy pile based on spatial-temporal feature extraction through CNN-LSTM hybrid model

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File organization: the next page begins the author-accepted manuscript. Appendix A follows the main manuscript references.

# Outlet water temperature prediction of energy pile based on spatial-temporal feature extraction through CNN–LSTM hybrid model

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## Abstract

Energy pile is a novel ground heat exchanger for ground source heat pump (GSHP) systems. Prediction of the energy pile outlet water temperature is essential for the efficient operation of GSHP systems. In this study, by establishing a convolutional neural network (CNN) and long short-term memory (LSTM) hybrid model (CNN-LSTM), the spatial-temporal feature of the soil temperature field (STF) was creatively considered to predict the outlet water temperature. The inlet and outlet water temperatures and the surrounding STF data of the energy pile were obtained through finite element simulation and used as the model training datasets. By building the CNN-LSTM model, the spatial-temporal features in datasets could be extracted, leading to more accurate prediction results than other benchmark models. For instance, the excellent prediction accuracy of CNN-LSTM is reflected by an average  $R^2$  value of 96.252%, which is higher than the values of the LSTM, CNN, ANN, and SimpleRNN models by 2.326%, 3.527%, 4.585%, and 5.755%, respectively. Furthermore, the influence of different STF datasets on the prediction accuracy was investigated. The corresponding dataset acquisition method based on the optimized sensor arrangement scheme was proposed, which can improve the information extraction performance of the CNN-LSTM model.

**Keywords:**

Energy pile; Ground source heat pump; Outlet water temperature prediction; Deep learning; Spatial-temporal feature; Soil temperature field.

**Abbreviations**

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| GSHP           | Ground source heat pump                                              |
| CNN            | Convolutional neural network                                         |
| LSTM           | Long short-term memory                                               |
| ANN            | Artificial neural network                                            |
| RNN            | Recurrent neural network                                             |
| GHE            | Ground heat exchanger                                                |
| CNN-LSTM       | Convolutional neural network and long short-term memory hybrid model |
| STF            | Soil temperature field                                               |
| COP            | Coefficient of performance                                           |
| ARMA           | Autoregressive moving average                                        |
| ARIMA          | Autoregressive integrated moving average                             |
| RNN            | Recurrent neural network                                             |
| RMSE           | Root mean square error                                               |
| MAE            | Mean absolute error                                                  |
| MAPE           | Mean absolute percentage error                                       |
| R <sup>2</sup> | Coefficient of determination                                         |

**Nomenclature:**

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| $c$         | Specific heat capacity (J/kg °C)                                              |
| $\rho$      | Density (kg/m <sup>3</sup> )                                                  |
| $L$         | Water flow rate (m <sup>3</sup> /h)                                           |
| $l$         | Length of the energy pile (m)                                                 |
| $f_t$       | Forget gate at time $t$                                                       |
| $i_t$       | Input gate at time $t$                                                        |
| $o_t$       | Output gate at time $t$                                                       |
| $\sigma$    | Sigmoid activation function                                                   |
| $\tanh$     | Hyperbolic tangent functions                                                  |
| $Q$         | Heating capacity                                                              |
| $m_w$       | Mass flux of circulating water (kg/s)                                         |
| $c_w$       | Specific heat of circulating water (J/kg °C)                                  |
| $t_{out}$   | Outlet water temperature (°C)                                                 |
| $t_{in}$    | Inlet water temperature (°C)                                                  |
| $q$         | Heat exchange rate per metre (W/m)                                            |
| $x_t$       | Input data at time $t$                                                        |
| $h_t$       | Hidden state at time $t$                                                      |
| $n$         | Prediction range                                                              |
| $y_i$       | True value of the energy pile outlet water temperature at a certain time      |
| $\hat{y}_i$ | Predicted value of the energy pile outlet water temperature at a certain time |

## 1. Introduction

According to the statistical data of the International Energy Agency, building operations accounted for 30% of the global total final energy consumption and 28% of energy-related CO<sub>2</sub> emissions in 2019 [1]. To solve these problems, developing renewable energies, improving energy efficiency, and promoting their synergies are the core solutions [2].

Geothermal energy is one of the most widely used renewable energies for heating and cooling in the building sector. Compared with other renewable energy sources, geothermal energy is less affected by seasonal variations, has exceptionally consistent availability, and can be easily integrated with other energy sources [3]. It can be extracted by ground source heat pump (GSHP) systems through ground heat exchangers (GHEs), and GSHP systems have the largest share of worldwide annual direct geothermal use in 2020 (58.8%) [4]. Energy pile, as a relatively novel GHE for GSHP systems, integrates heat exchange loops with the building pile foundation, enabling significant land reduction and cost savings [5] for the initial investment of GSHP systems.

For daily usage, due to the thermal inertia of the ground and buildings, there is a time delay between the adjustment in the operating state of GSHP systems and the corresponding indoor temperature change [6]. Meanwhile, considering the uncertain fluctuations of the ambient temperature, it is very challenging to formulate short-term control strategies for GSHPs. Thus, reliable and efficient performance prediction of GSHPs is essential for optimizing control strategies. Because performance prediction of GSHPs can help formulate a model predictive control strategy and make control decisions in advance, the energy consumption can be reduced, and the indoor thermal comfort can be improved [7].

With the development and widespread application of smart sensors in recent years, using operation monitoring data to predict the performance of GSHP systems is becoming a research focus. The parameters for prediction include but are not limited to the coefficient of performance (COP) of GSHP systems [8], the heating capacity of GHEs [9], and the outlet temperature of GHEs [10, 11]. This study took GSHP systems with energy piles in heating mode as the research subject and focused on short-term outlet water temperature prediction.

The operation data of energy piles in daily use are time series data, which can be

collected in real time. Several classical statistical models, such as autoregressive moving average (ARMA) [12] and autoregressive integrated moving average (ARIMA) [13], have been widely used for time series prediction. However, these statistical models usually rely on the stationarity assumption [14] and can hardly capture complex interactions from nonlinear data [15]. When there is a sudden change in ambient temperature, the monitoring data of energy piles tend to violate the stationarity assumption, leading time series models such as ARMA and ARIMA to be unable to accurately predict the short-term performance of energy piles.

Machine learning models work as a black box and can find complex relations between nonlinear input and output characteristics [16]. Currently, most studies on GSHP performance prediction rely on shallow machine learning models, such as artificial neural networks (ANNs) and random forests. By building ANN models, Fannou et al. [9] predicted the heating capacity and power consumption of GSHPs based on experimental monitoring data. Gang et al. [10, 11] set the outlet water temperature of the GSHP as the prediction goal. Although these studies show that black box models are feasible for predicting the performance of GSHPs, the influence of the soil temperature field (STF) on prediction accuracy has been ignored. The STF varies with time and space, which strongly influences GHE performance [17]. During the operation of a GSHP system, the temperature load can lead to thermal accumulation in the soil, causing a decline in heating efficiency [18, 19]. Therefore, the STF should be included in the datasets to help networks obtain more information, thereby improving prediction accuracy. Lu et al. [8] built random forest models to predict the COP of GSHP systems and investigated the influence of different input parameters on prediction accuracy. Their results proved that soil temperature is one of the most important parameters that affect prediction accuracy. Nevertheless, for soil temperature, their study only took the monitoring data at a vertical depth of 2.5 m in the soil as the input variable. The temperature from one location cannot fully represent the STF and its influence on an energy pile in space. Therefore, to use the soil temperature data in the prediction of energy pile performance, it is more comprehensive to consider the spatial information of the STF. However, the conventional shallow machine learning models mentioned above can only extract superficial features [20] and ignore deep relationships, such as soil temperature spatial features from operational monitoring data.

Compared to classical statistical models and shallow machine learning models, deep learning models use multilayer networks to extract more features from the original input data [21]. In recent years, deep learning models, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), have been extensively studied in various scientific fields. The long short-term memory (LSTM) network, one RNN variant, can handle long-term dependencies implied in time series well [22]. Xie et al. [23] used the excellent time-series processing ability of LSTM, established TRNSYS models of a GSHP system to collect datasets, and predicted the hour-ahead cooling load of the GSHP system. In 2015, Donahue et al. [24] first proposed a long-term recurrent convolutional network for visual recognition and description; in this model, a CNN was used as a front end of LSTM. This type of hybrid model is now known as the CNN-LSTM model, and it has been used in many fields, such as in predicting building heat load [25] and air quality index [26] and in photovoltaic plant power production [27], in which datasets are spatiotemporally related.

To the knowledge of the authors, the applicability of the CNN-LSTM hybrid model has never been investigated in the energy pile performance prediction area. This study aims to fill the knowledge gaps in how to achieve efficient and accurate prediction of energy pile outlet water temperature by exploiting the multifeatured extraction capability of the hybrid deep learning model and constructing reasonable datasets that can improve the accuracy of the model. In this study, the datasets used for model training were generated from validated finite-element models, and the prediction results of the established CNN-LSTM hybrid model were compared to those of other benchmark models to verify its advantages. Furthermore, the influence of different STF datasets on the prediction accuracy was specifically investigated, and the corresponding optimal sensor arrangement scheme was proposed.

## **2. Methodology**

The methodologies used in this research are shown in Fig. 1. First, based on the verified finite element model, the sequences of the monitoring data were generated to build datasets. The datasets included the soil temperature and the water temperatures at the inlet and the outlet of the energy pile. Two different working conditions of the energy pile were considered, i.e., with and without the groundwater flow influence (Example 1 and Example 2). Second, the CNN-LSTM model was established and

verified, and further studies were conducted to reveal the influence of the STF on accuracy prediction.

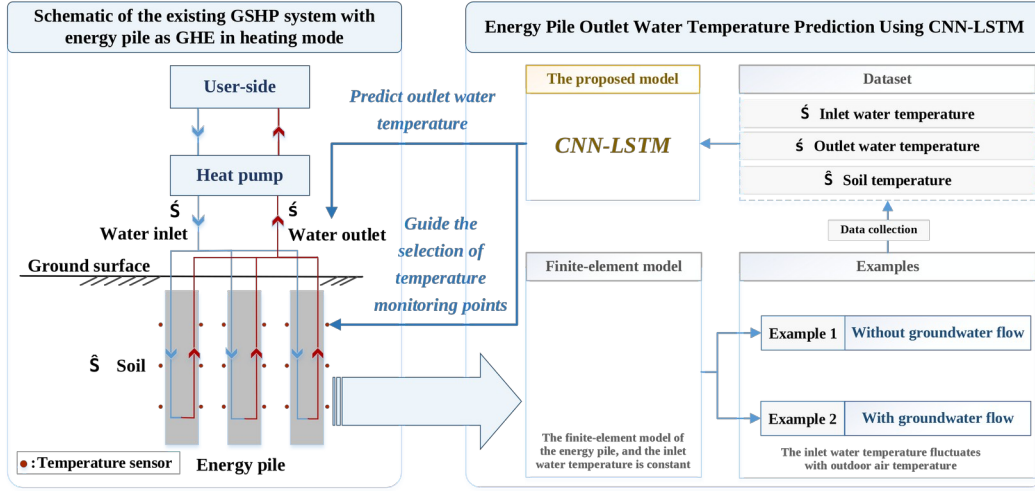


Fig. 1. Schematic diagram of the research methodology.

## 2.1. Finite element model

This study utilized finite-element simulation to generate the network training datasets. The finite-element model was built based on the in situ energy pile performance experiments conducted by Gao et al. [28, 29] in Shanghai. In the in situ test [28], the diameter of the pile was 0.6 m, and the length of the pile was 25 m. The pile was laid at a depth of 10.4 m, where the soil temperature remained constant at 18.2 °C. The thermal medium was water, which flowed in the W-shaped pipes and was cased in the piles.

The 3D geometric models of the energy pile and soil were developed using finite-element analysis software (COMSOL Multiphysics). The geometric dimensions of the finite-element model are illustrated in Fig. 2. As shown in Fig. 2, the diameter and length of the pile were kept the same as those in the experiment, and the computational domain of the soil was set to 4 m\*4 m\*27 m.

The mesh of the finite-element model was generated with a tetrahedral configuration, and the mesh near the pipes was refined to ensure simulation accuracy. Grid sensitivity analyses were conducted to investigate the influence of tetrahedral mesh elements with different element sizes on the accuracy of the numerical model



results. Specifically, four different sets of meshes with different grid numbers (145385, 216197, 260986, and 293761) were divided for the same numerical model. The heat exchange simulations of the energy pile were conducted with different grid numbers. The simulated outlet water temperature is shown in Table 1. The results reveal that the simulation results from grid schemes 2~4 are almost the same and are closer to the experimental value than those of grid scheme 1. Therefore, this study used grid scheme 2 to achieve an accurate result with lower computational cost.

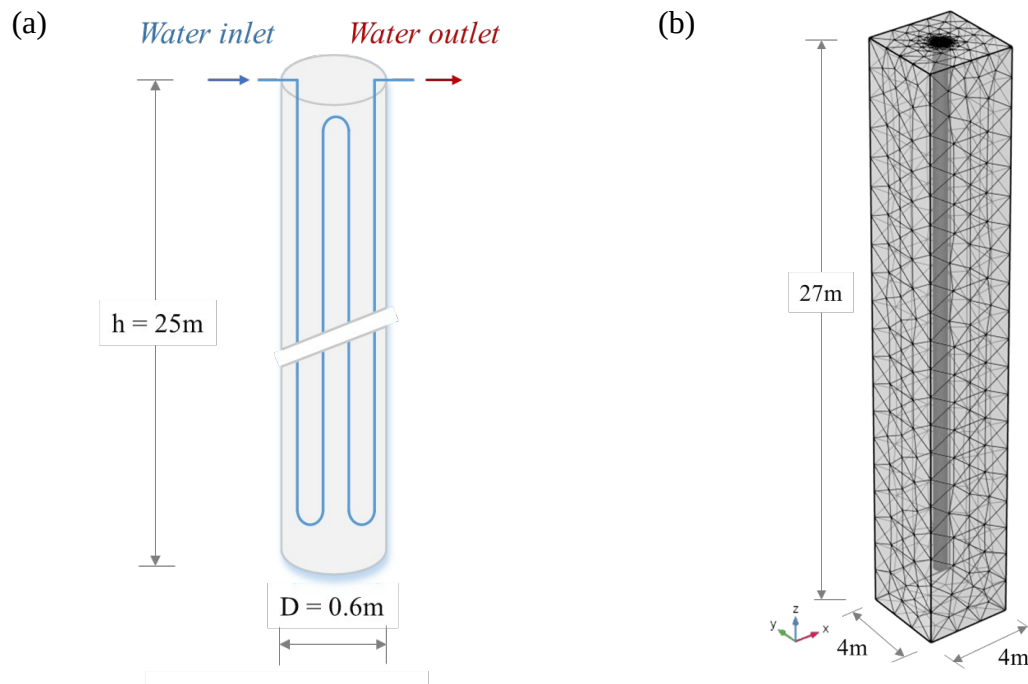


Fig. 2. Geometric dimensions of the finite-element model: (a) geometry of the pile and W-shaped pipe in the pile and (b) finite element mesh and soil dimensions.

**Table 1**

Outlet water temperature of the energy pile at 420 min with different grid schemes.

|                                                    |              | Grid scheme 1 | Grid scheme 2 | Grid scheme 3 | Grid scheme 4 |
|----------------------------------------------------|--------------|---------------|---------------|---------------|---------------|
| Grid number                                        |              | 145385        | 216197        | 260986        | 293761        |
| Outlet temperature<br>of the W-shaped<br>pile (°C) | Simulation   | 29.21         | 29.27         | 29.27         | 29.27         |
|                                                    | Experimental |               | 29.88         |               |               |

Verification of the finite-element model was conducted by comparing the results

of heat exchange simulations for the energy pile with the in situ test measurement data [28, 29]. Two simulations with different flow rates, 0.342 and  $0.342 \times 2 \text{ m}^3/\text{h}$ , in W-shaped pipes were considered.

The material constants and boundary conditions set in the simulation are presented in Table 2 and Table 3, respectively. The heat exchange rate per metre of the energy pile was used as the evaluation index of the energy pile performance [29]. It is described as shown in Equation (1):

$$q = \frac{c\rho L (t_{in} - t_{out})}{l}, \quad (1)$$

where  $q$  is the heat exchange rate per metre (W/m);  $c$  is the specific heat capacity (J/kg °C),  $\rho$  is the density (kg/m<sup>3</sup>),  $L$  is the water flow rate (m<sup>3</sup>/h),  $t_{in}$  is the inlet water temperature (°C),  $t_{out}$  is the outlet water temperature (°C), and  $l$  is the length of the energy pile (m).

**Table 2**

Physical properties of the materials [28] used in this study.

| Material                                        | Thermal conductivity<br>(W/m °C) | Density (kg/m <sup>3</sup> ) | Heat capacity (J/kg<br>°C) |
|-------------------------------------------------|----------------------------------|------------------------------|----------------------------|
| Soil                                            | 1.3                              | 1847                         | 1200                       |
| Concrete                                        | 1.628                            | 2500                         | 837                        |
| Pipe (pipe inner diameter, $d = 2 \text{ cm}$ ) | 0.42                             | \                            | \                          |

**Table 3**

Boundary conditions and experimental parameters of the model

| Boundary conditions             | Description                                                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Inlet velocity                  | 0.3 m/s and 0.6 m/s (inlet velocity = $4L/\pi d^2$ )                                                                             |
| Inlet temperature               | 35.02 °C                                                                                                                         |
| Outlet                          | Single direction outflow                                                                                                         |
| Soil boundary                   | The vertical surfaces and the top surface of the soil zone are assumed to be adiabatic. The bottom surface is set to be 18.2 °C. |
| Initial temperature of the soil | 18.2 °C                                                                                                                          |

In this study, the specific heat capacity of the circulating water is 4185 J/kg °C, the

density is  $998 \text{ kg/m}^3$ , and the length of the energy pile is 25 m. According to Equation (1), the comparison of the simulation result ( $q$ ) with the experimental result is shown in Table 4. The simulated outlet water temperature and corresponding calculated  $q$  are quite close to those of the in situ experiments. Therefore, the finite-element model is verified to be accurate. By using the validated finite-element model, this research changed the environmental conditions to generate different datasets, and the specific parameter settings are detailed in Section 3.2.

**Table 4**

Comparison of the simulation result ( $q$ ) with the experimental result

| Types                                                                                                     |              | Inlet water temperature<br>$t_{in} \text{ (}^\circ\text{C)}$ | Water flow rate<br>$L \text{ (m}^3\text{/h)}$ | Outlet water temperature<br>$t_{out} \text{ (}^\circ\text{C)}$ | Heat exchange rate per metre $q$<br>$\text{(W/m)}$ |
|-----------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------|----------------------------------------------------|
| W-shaped pile                                                                                             | Experimental | 35.02                                                        | 0.342                                         | 29.88                                                          | 83.05                                              |
|                                                                                                           | Simulation   | 35.02                                                        | 0.342                                         | 29.27                                                          | 92.91                                              |
|                                                                                                           | % Error      |                                                              |                                               | 2.04%                                                          | 11.87%                                             |
| W-shaped pile with the double flow rate                                                                   | Experimental | 34.79                                                        | $0.342 \times 2$                              | 31.88                                                          | 94.25                                              |
|                                                                                                           | Simulation   | 34.79                                                        | $0.342 \times 2$                              | 31.67                                                          | 98.18                                              |
|                                                                                                           | % Error      |                                                              |                                               | 0.66%                                                          | 4.00%                                              |
| $\% \text{ Error} = \frac{ \text{Simulation} - \text{Experimental} }{ \text{Experimental} } \times 100\%$ |              |                                                              |                                               |                                                                |                                                    |

## 2.2. CNN-LSTM hybrid model

A CNN-LSTM hybrid model was established, and the structure and the data processing flow of the proposed CNN-LSTM hybrid model are shown in Fig. 3. The CNN layer primarily extracts the spatial features from the monitoring data. The obtained features are then flattened into a one-dimensional array and used as the input for the LSTM layer to analyse the time-series features. The feature processing mechanism of the CNN and LSTM is shown in the following subsections.



LSTM hybrid model included  $t_{in}$ ,  $t_{out}$ , and soil temperature data. Therefore, the feature extraction of the CNN layer in the hybrid model works as follows. First, the input layer of the CNN receives input data from the dataset. Then, the convolutional layer uses kernels to extract among between  $t_{in}$ ,  $t_{out}$ , and soil temperature data. Through this operation, the spatial features of the STF data can be effectively extracted. After this, the max-pooling layer works to compress the amount of data from the convolutional layer, reducing overfitting [31]. Finally, the flattened layer reduces the dimension of the parameters, and the parameters can be input to the LSTM layer to process the time series.

### 2.2.2. Time series processing based on LSTM

In this study, LSTM is used to process the time series data after spatial feature extraction by the CNN. The LSTM network is a type of RNN. An RNN uses sequence data as input and consists of a series of modules with the same basic structure. In an RNN, the connections between modules form a directed cycle. However, an RNN cannot transmit information for long periods. When a time series becomes too large, the accuracy of an RNN model decreases [32]. Hochreiter and Schmidhuber [33, 34] proposed an LSTM network to solve this problem. Compared to the traditional RNN, the LSTM network has more complex unit states. Its internal memory unit and gate mechanism can overcome the gradient disappearance and gradient explosion problems in traditional RNN training [35]. The network structure of LSTM is presented in Fig. 3. The core calculation steps of the LSTM model are presented in Equations (5) to (10).

The equation for the forget gate  $f$  is shown as Equation (5). It decides what information should be discarded from the cell state based on the hidden state  $h_{t-1}$  and input data  $x_t$ :

$$f_t = \sigma ( w_f \cdot [ h_{t-1}, x_t ] + b_f ) \quad (5)$$

where  $\sigma$  is the sigmoid activation function.  $w$  and  $b$  represent the weight matrix and bias, respectively.  $h$  is the hidden state.  $x$  is the input data.

The equations for the input gate  $i$  is

$$i_t = \sigma ( w_i \cdot [ h_{t-1}, x_t ] + b_i ) \quad (6)$$

$$\tilde{c}_t = \tanh ( w_c \cdot [ h_{t-1}, x_t ] + b_c ) \quad (7)$$

where  $\tilde{c}_t$  is the process inputs of neurons at time  $t$  and  $w_c$  and  $b_c$  represent the weight and bias for the input data of the neurons in the hidden layer when it switches from time  $t-1$

to time  $t$ .

The updated intermediate state variable  $c_t$  at the current time is

$$c_t = f_t ( w_c \cdot c_{t-1} + i_t \cdot \tilde{c}_t ) \quad (8)$$

The equation updates the cell state of the present time step  $c_t$ , which combines the candidate memory  $\tilde{c}_t$  and the long-term memory  $c_{t-1}$ .

The output gate  $O$  can be expressed as

$$o_t = \sigma ( w_o \cdot [ h_{t-1}, x_t ] + b_o ) \quad (9)$$

Thus, the output of the current neurons is

$$h_t = o_t \cdot \tanh ( c_t ) \quad (10)$$

The above formulas show that LSTM can handle continuous and highly correlated time series data. During the heat exchange operation of energy pile systems, the current temperature monitoring data are closely related to the previous moment, and the series of monitoring data is highly time-dependent. Therefore, LSTM, with a powerful correlation and feedback mechanism, can process the time series information in the parameters inputted from the CNN and send the processed data to the fully connected layer to output the predicted outlet water temperature.

### 2.3. Evaluation of model performances

In this study, as shown in Fig. 4, ANN, CNN, SimpleRNN, and LSTM models were used as the benchmark models to evaluate the prediction performance of CNN-LSTM. These four models were chosen as the benchmark models because they are classical neural network models and have been applied in numerous studies in the field of GSHP prediction [8-11]. Moreover, the CNN and LSTM are the components of the proposed CNN-LSTM hybrid model.

Four indices were used to evaluate the accuracy of the prediction models, the root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination ( $R^2$ ), as given by Equations (11) to (14):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n ( y_i - \hat{y}_i )^2}, \quad (11)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n | \hat{y}_i - y_i |, \quad (12)$$

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i}, \text{ and} \quad (13)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2}, \quad (14)$$

where  $n$  represents the prediction range and  $y_i$  and  $\hat{y}_i$  represent the true value and predicted value of the outlet water temperature of the energy pile at a certain time, respectively.

To ensure the reliability of each neural network and the accuracy of the evaluation parameters, in the following sections, unless otherwise specified, the calculated RMSE, MAE, MAPE, and  $R^2$  are the average values of the optimal structure of each network verified by multiple experiments after ten consecutive runs.

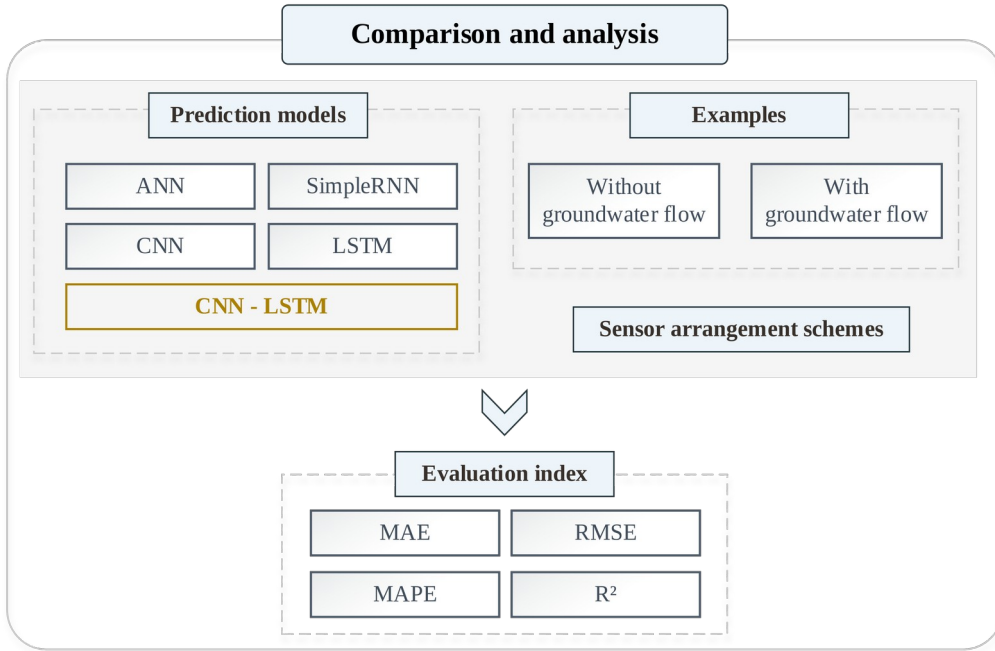


Fig. 4. Performance evaluation index of the predictive model.

### 3. Data preparation and pre-processing

#### 3.1. Parameters for prediction

##### 3.1.1. Inlet and outlet water temperatures of the energy pile

This study takes the heating mode of an energy pile system operating in winter as an example. The heating capacity  $Q$  of the energy pile system corresponds to the energy

extracted from the soil, which can be calculated by Equation (15):

$$Q = m_w c_w (t_{out} - t_{in}), \quad (15)$$

where  $m_w$  is the mass flux of circulating water (kg/s),  $c_w$  is the specific heat of the circulating water (J/kg °C),  $t_{out}$  is the outlet water temperature (°C), and  $t_{in}$  is the inlet water temperature (°C).

Generally, for a specific energy pile system, the type of circulating fluid is fixed, so  $c_w$  is constant and  $m_w$  can be adjusted by controlling the fluid flow rate.  $t_{in}$  can be collected by monitoring the inlet water temperature. Therefore, the outlet water temperature  $t_{out}$  is a direct indicator of the heating performance of the energy pile. The prediction of  $t_{out}$  can assist in the optimal control and intelligent adjustment of energy pile systems [10, 11]. Thus, this study focused on predicting the outlet water temperature  $t_{out}$ .

In practice, the inlet water temperature  $t_{in}$  varies with the fluctuating thermal loads of a building, so the fluctuating inlet water temperature should be used when performing numerical simulations for the energy pile system in this study. However, in the in situ experiments of references [28, 29], the inlet water temperature of the energy pile system was set to three different constant temperatures of 32 °C, 35 °C, and 38 °C. Obviously, the constant inlet water temperature cannot reflect the continuous fluctuation in the thermal load of buildings. As we did not have the in situ fluctuating inlet water temperature of the energy pile system, we used the recorded outdoor air temperature of Shanghai to represent it as an alternative. To validate the correctness of this alternative, the Pearson correlation coefficient [36] for the outdoor air temperature and the inlet water temperature was calculated (see Appendix A for details), and the temperature data were obtained from the experiments of Esen et al. [37]. There is a strong linear correlation between the inlet water temperature and the outdoor air temperature. In fact, when the outdoor air temperature changes, the indoor thermal load also changes accordingly, which in turn causes the inlet water temperature of the energy pile to change. Therefore, the inlet water temperature of the energy pile is heavily influenced by the outdoor air temperature, and it is feasible to use the outdoor air temperature as an alternative for the energy pile inlet water temperature to train the models in this study.

Thus, in the simulations, the outdoor air temperature of Shanghai for the period



from January 09, 2020, to January 23, 2020 [38], was used as an alternative to the inlet water temperature for the energy pile. The temperature data were recorded every 30 min.

### **3.1.2. Soil temperature field data**

Soil temperature is a significant parameter for energy pile outlet water temperature prediction. In this study, temperature data from 12 locations in the soil,  $t_{in}$  and  $t_{out}$ , for a total of 14 energy pile features for the monitoring data, were included to build datasets for the prediction model. These temperature data generated from the finite-element simulation can also be obtained by arranging the corresponding sensors in the actual project.

Additionally, if groundwater flow exists in the soil, it affects the performance of the GSHP [39-41]. Therefore, to investigate the prediction accuracy of the models in different underground environments, datasets from different simulation conditions (with and without groundwater flow) were used for model training. The different underground environments built in this research are divided into two examples. As shown in Fig. 5, Example 1 does not consider groundwater flow influence, while Example 2 considers groundwater flow influence. The soil in Example 2 was affected by the flow of groundwater, and to reflect the influence of groundwater on the STF, the groundwater velocity was set to  $10^{-5}$  m/s [42]. To monitor the temperature evolution in the soil during the simulation, 12 temperature monitoring sensors were set at a distance 0.5 times the pile diameter along the horizontal direction of the pile [43]. The distribution positions of the sensors along the longitudinal direction of the pile are shown in Fig. 5.

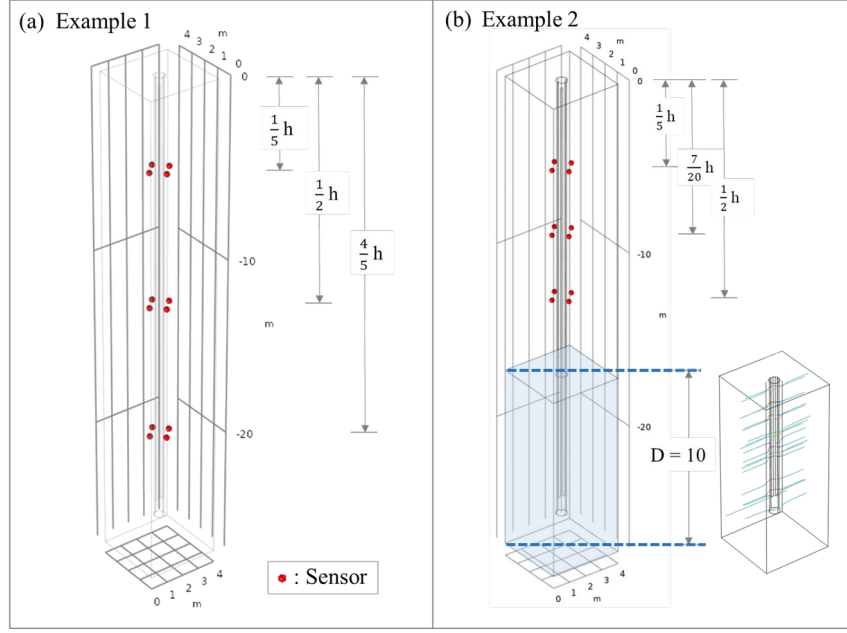


Fig. 5. Schematic of the finite-element model. (a) Example 1: Without groundwater flow. (b) Example 2: With groundwater flow.

### 3.2. Datasets and data pre-processing

In the finite-element simulation, when cold water was first introduced into the energy pile, the soil did not reach a stable heat transfer state. Therefore, the stable simulation results from January 12, 2020, to January 23, 2020, were used to construct the dataset. Each dataset contained 14 digital features ( $t_{in}$ ,  $t_{out}$ , and temperature data from 12 sensors in the soil). For each dataset, the first 80% were used to train the model, while the last 20% were used to test the accuracy of the model. The data from the previous hour was used to predict the data for the next 30 min. Figure 6 shows the inlet and outlet water temperature datasets for Example 1. There were 458 samples in the training set and 115 samples in the test sets. Figure 7 shows the dataset comparison between Example 1 and Example 2. As the flow of groundwater can enhance the heat exchange performance of the soil [44], the outlet water temperature of Example 2 is higher than that of Example 1.

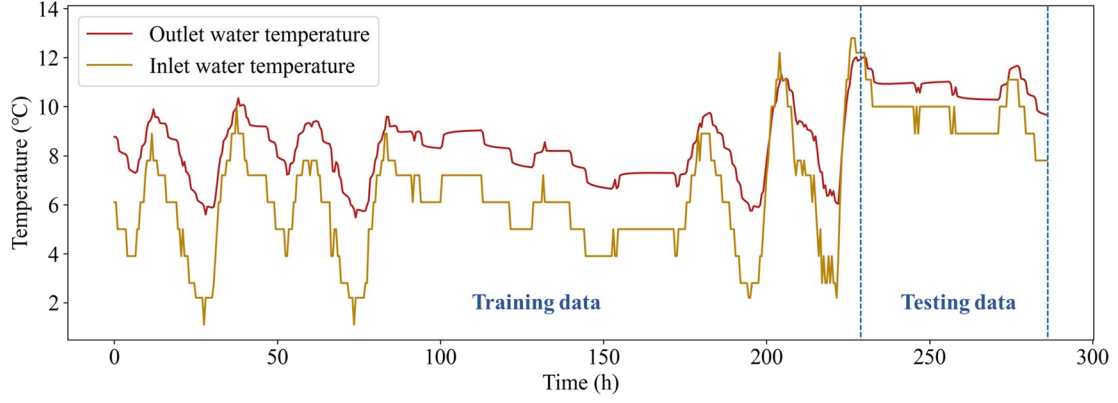


Fig. 6. Inlet and outlet water temperature datasets in Example 1.

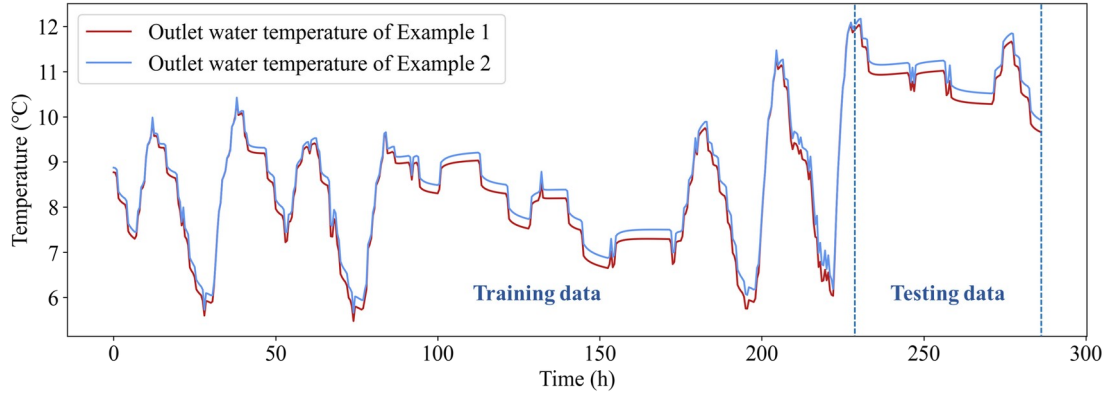


Fig. 7. Outlet water temperature dataset comparison between Example 1 and Example 2.

## 4. Results and discussion

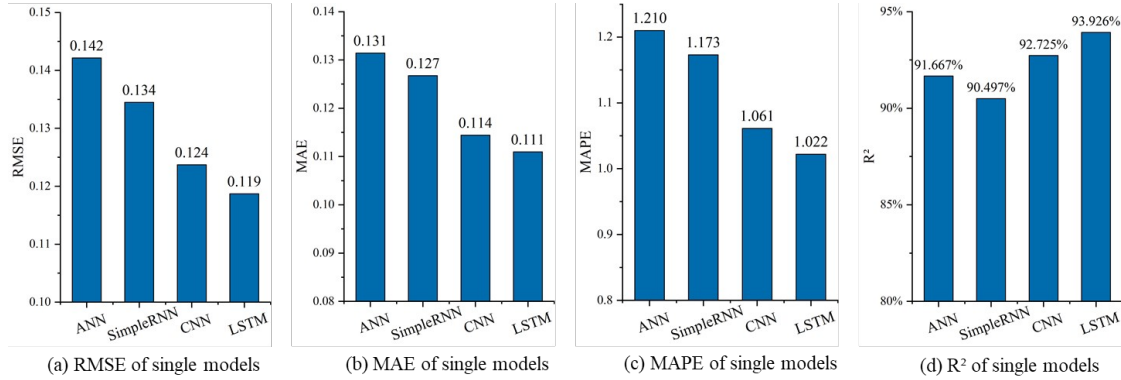
### 4.1. Prediction performance of benchmark models with STF data

To verify the superiority of the established CNN-LSTM model, this study constructed ANN, SimpleRNN, CNN, and LSTM models as the benchmark models and used the data generated in Example 1 as the dataset for model training. Table 5 lists the optimal network structures of the four benchmark models, as determined by numerous tests. This study used padding [45] to avoid information loss when a CNN was used for feature extraction. The prediction capabilities of benchmark models were compared. The prediction accuracy was evaluated by the evaluation indicators RMSE, MAE, MAPE, and  $R^2$ . Fig. 8 shows the comparison results of the evaluation indicators of the four models. In Example 1, the CNN and LSTM achieve better performance than the ANN and SimpleRNN.

**Table 5**

Optimal neural network structures of benchmark models

| Layers                        | Models     |            |            |                                                            |
|-------------------------------|------------|------------|------------|------------------------------------------------------------|
|                               | ANN        | SimpleRNN  | LSTM       | CNN                                                        |
| 1                             | Dense = 64 | Units = 64 | Units = 64 | Filters = 64, kernel_size = (1,4)<br>MaxPooling2D = (1, 4) |
| 2                             | Dense = 64 | Units = 64 | Units = 64 | Filters = 64, kernel_size = (1,4)<br>MaxPooling2D = (1,2)  |
| 3                             | Dense = 1  | Dense = 1  | Dense = 1  | Dense = 64                                                 |
| 4                             |            |            |            | Dense = 64                                                 |
| 5                             |            |            |            | Dense = 1                                                  |
| Epochs = 100, batch_size = 16 |            |            |            |                                                            |

**Fig. 8.** RMSE, MAE, MAPE, and R<sup>2</sup> comparison results of the benchmark models.

#### 4.2. Prediction performance of CNN-LSTM with the STF data

A CNN-LSTM hybrid model was established as described in Section 2.2 and as shown in Table 6, and this CNN-LSTM model was used to extract the temporal features and to process the time series features of the dataset simultaneously.

**Table 6**

Established CNN-LSTM architecture

| Layer (type) | Structure                         | Parameter |
|--------------|-----------------------------------|-----------|
| Conv2D       | Filters = 96, kernel_size = (1,2) | 288       |

|                               |                                    |        |
|-------------------------------|------------------------------------|--------|
|                               | MaxPooling2D = (1,2)               | 0      |
| Conv2D                        | Filters = 192, kernel_size = (1,2) | 37056  |
|                               | MaxPooling2D = (1,2)               | 0      |
|                               | Flatten                            | 0      |
|                               | RepeatVector                       | 0      |
| LSTM                          | Units = 64                         | 311552 |
| LSTM                          | Units = 64                         | 33024  |
| Dense                         | Dense = 1                          | 65     |
| Epochs = 100, batch_size = 16 |                                    |        |

Based on the average prediction results of training the CNN-LSTM network structure shown in Table 6 ten consecutive times, the RMSE is 0.096, the MAE is 0.078, the MAPE is 0.723, and  $R^2$  is 96.252%. The  $R^2$  value is higher than the  $R^2$  values of the LSTM, CNN, ANN, and SimpleRNN models by 2.326%, 3.527%, 4.585%, and 5.755%, respectively, as shown in Fig. 8. To provide a more intuitive comparison, the temperature curve predicted for each neural network when the evaluation index was close to the average value is shown in Fig. 9. The outlet water temperature prediction results of CNN-LSTM ( $R^2 = 96.516\%$ ) are better than those of LSTM ( $R^2 = 93.519\%$ ), the CNN ( $R^2 = 92.460\%$ ), the ANN ( $R^2 = 91.845\%$ ), and SimpleRNN ( $R^2 = 90.557\%$ ). Based on the above results, without the influence of groundwater flow, the CNN-LSTM hybrid model exhibited better prediction performances than the other benchmark models.

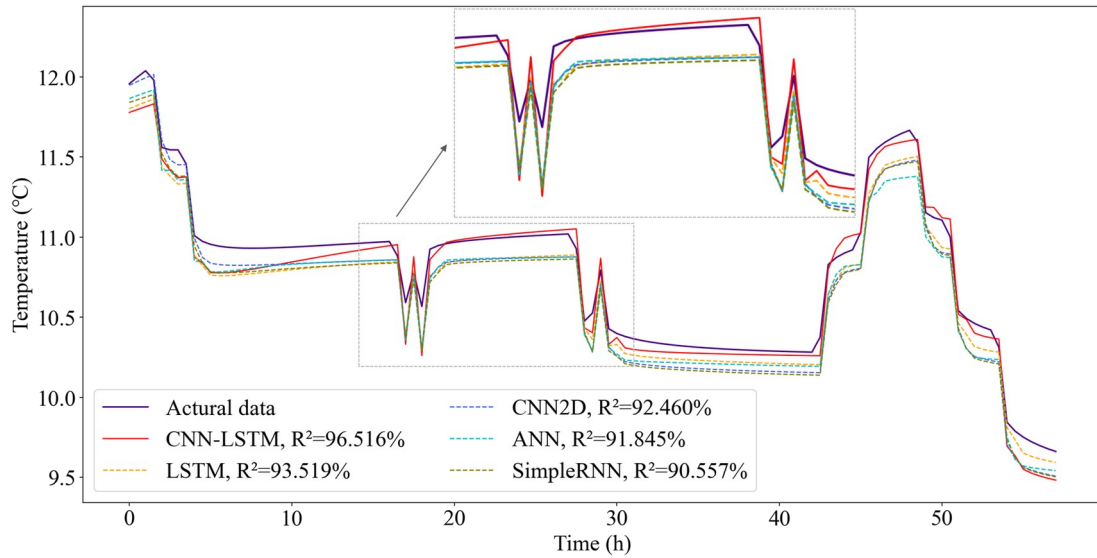


Fig. 9. Prediction comparison of the CNN-LSTM hybrid model and benchmark

models.

#### 4.3. Influence of STF data on the prediction accuracy of CNN, LSTM, and CNN-LSTM models

To quantitatively study how the STF data improve the prediction accuracy of deep learning models, CNN-LSTM, CNN, and LSTM models were trained using the dataset without STF data ( $t_{in}$  and  $t_{out}$ ) and with STF data ( $t_{in}$ ,  $t_{out}$  and temperature data of 12 sensors in the soil). The datasets were acquired based on the simulation results of Example 1, in which the influence of groundwater on the energy pile is not considered (as shown in Fig. 5 (a)). The accuracies of the prediction results were compared, as shown in Table 7. When considering the STF data, the  $R^2$  values of the results of the CNN-LSTM, CNN, and LSTM models are 3.53%, 0.893%, and 3.19% higher than those without STF data, respectively. This demonstrates that when the STF data were considered in the dataset, the prediction accuracies of the CNN, LSTM, and CNN-LSTM hybrid models could all be improved.

**Table 7**

Comparison of CNN-LSTM performance without and with STF data

|       | Without STF data |         |         | With STF data |         |         |
|-------|------------------|---------|---------|---------------|---------|---------|
|       | CNN-LSTM         | CNN     | LSTM    | CNN-LSTM      | CNN     | LSTM    |
| RMSE  | 0.118            | 0.130   | 0.146   | 0.096         | 0.124   | 0.119   |
| MAE   | 0.109            | 0.121   | 0.138   | 0.078         | 0.114   | 0.111   |
| MAPE  | 1.012            | 1.121   | 1.272   | 0.723         | 1.061   | 1.022   |
| $R^2$ | 92.722%          | 91.832% | 90.736% | 96.252%       | 92.725% | 93.926% |

#### 4.4. Prediction results with the influence of groundwater flow on the STF

To analyse the applicability of the CNN-LSTM hybrid model under different underground environmental conditions, CNN, LSTM, and CNN-LSTM prediction models under groundwater flow influence (Example 2) were developed. As shown in Table 8, compared to the CNN-LSTM used for Example 1 (Table 6), the neural network structure of the CNN-LSTM (Example 2) hybrid model was adjusted appropriately in the CNN layer, with the LSTM layer kept the same as the one used in the benchmark model. The dataset input to the network was changed to the finite element simulation results of Example 2.

**Table 8**

Neural network structure of the CNN, LSTM, and CNN-LSTM models for Example 2

| Layer | CNN                                                        | LSTM       | CNN-LSTM                                                   |
|-------|------------------------------------------------------------|------------|------------------------------------------------------------|
| 1     | Filters = 64, kernel_size = (2,4)<br>MaxPooling2D = (1,2)  | Units = 64 | Filters = 64, kernel_size = (2,4)<br>MaxPooling2D = (1,2)  |
| 2     | Filters = 128, kernel_size = (2,4)<br>MaxPooling2D = (1,2) | Units = 64 | Filters = 128, kernel_size = (2,4)<br>MaxPooling2D = (1,2) |
| 3     | Dense = 64                                                 | Dense = 1  | LSTM units = 64                                            |
| 4     | Dense = 64                                                 |            | LSTM units = 64                                            |
| 5     | Dense = 1                                                  |            | Dense = 1                                                  |

Ten consecutive predictions were conducted through the CNN-LSTM (Example 1), CNN (Example 2), LSTM (Example 2), and CNN-LSTM (Example 2) models. Boxplots of the RMSE, MAE, MAPE, and  $R^2$  of the prediction results are shown in Fig. 10. In the boxplot shown in Fig. 10, five parameters describe a continuous training result for a neural network. The black horizontal line indicates the maximum and minimum values of the result; the yellow box represents the range between the 25th percentile and 75th percentile for the result; the blue horizontal line indicates the median value; the green triangle represents the mean value; and the hollow circle indicates the outliers of the data. Fig. 10 shows that in Example 2, the prediction performance of CNN-LSTM is more stable than that of the CNN and LSTM, with a narrower box and better prediction accuracy. Comparing the CNN-LSTM (Example 1) and CNN-LSTM (Example 2) boxplots shows that with the influence of groundwater flow, the box is wider, which means that the data are more dispersed and that the model is less stable. However, the average prediction accuracies of the CNN-LSTM model are similar in the two environments. Specifically, for  $R^2$ , with groundwater flow, the average value is 96.155% and is close to that without groundwater flow, which is 96.252%. Therefore, the CNN-LSTM hybrid model is more accurate and stable than other benchmark models for energy pile outlet water temperature prediction and is suitable for underground environments with or without underground water flow influence.

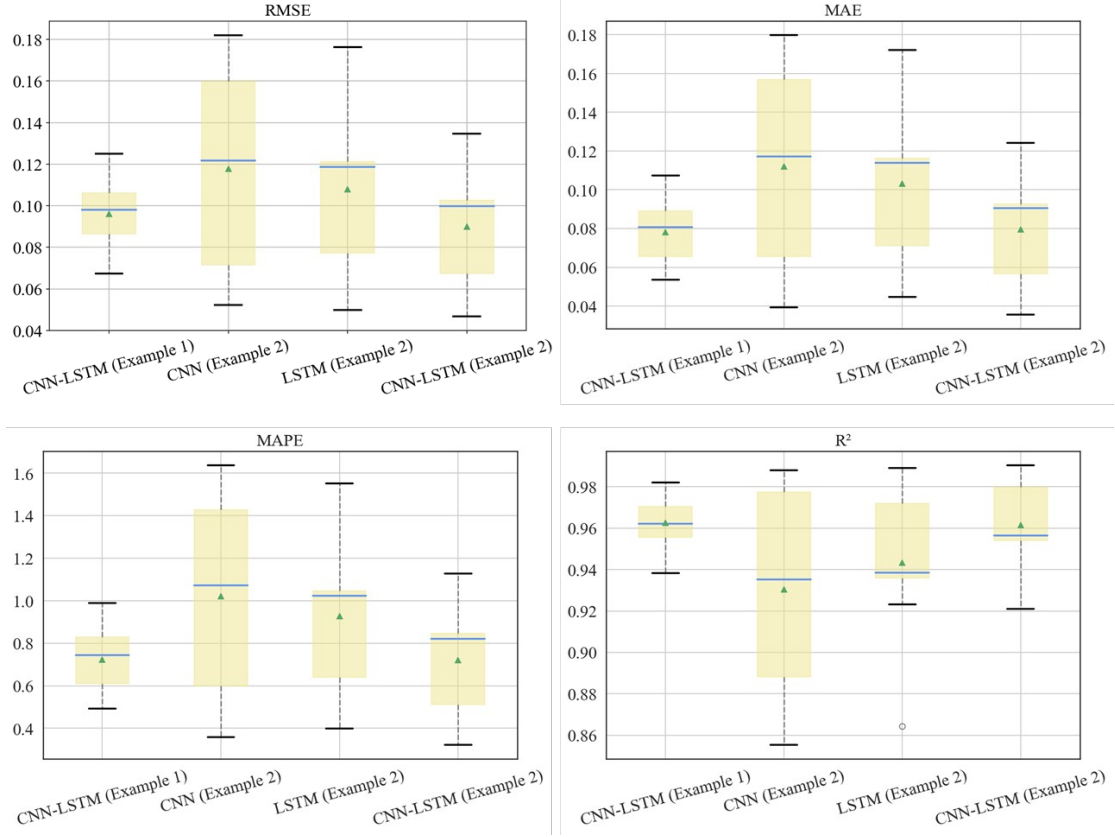


Fig. 10. Prediction accuracy comparison: RMSE, MAE, and MAPE boxplots for CNN-LSTM and LSTM in two examples.

#### 4.5. Influence of different STF datasets on the prediction accuracy

To further improve the prediction accuracy of CNN-LSTM, the dataset construction strategy can be optimized. In this study, by applying different soil temperature sensor arrangement schemes in the finite element simulation of Example 1 and Example 2, STF datasets with different spatial-temporal features were achieved, and the influence of different STF datasets on the prediction accuracy was investigated.

##### 4.5.1. Without groundwater flow

In Example 1, five different soil temperature sensor arrangement schemes are designed, namely, Example 1-1 to Example 1-5, as shown in Fig. 11. All sensors are located 0.5 times the pile diameter from the energy pile except for Example 1-5. In Examples 1-1, 1-2, and 1-3, the 12 sensors are arranged into 3 groups (4 sensors in each group) and distributed along the pile, and the distance between each group is shown in Fig. 11. In Example 1-4, all sensors are arranged on the same side of the energy pile, at the same distance from the water inlet and outlet, evenly distributed along the



longitudinal direction of the pile. In Example 1-5, the 12 sensors are evenly divided into three groups, and each group is evenly distributed in the circumferential direction with the energy pile as the axis. The distance between each group of sensors and the distance between the inner group sensors and energy pile are all half the diameter of the energy pile. The network structures of Examples 1-1 to 1-5 are the best network structures obtained by multiple training of the model.

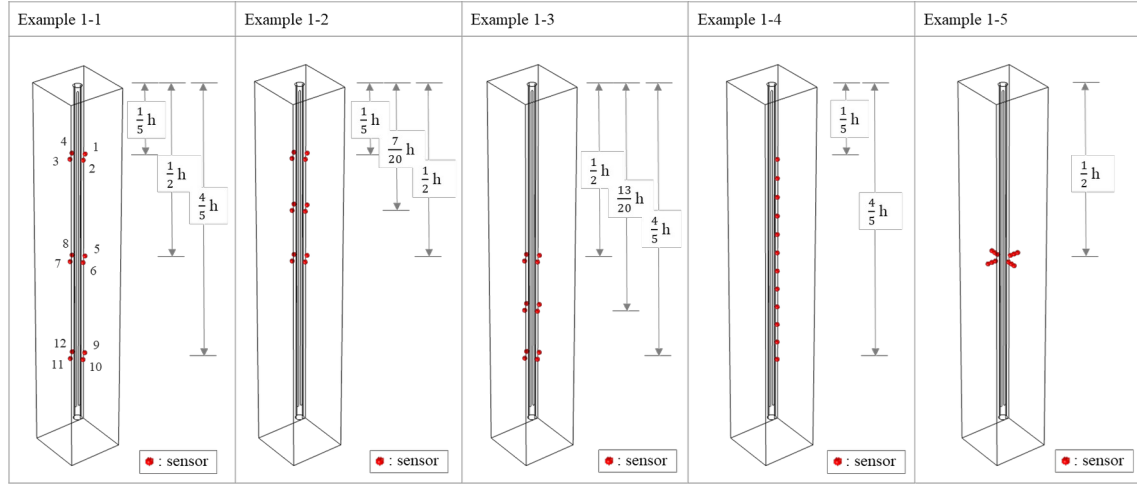


Fig. 11. Different sensor arrangement schemes in Example 1.

Table 9 shows the average RMSE, MAE, MAPE, and  $R^2$  values of the results obtained for each neural network after training 10 consecutive times. The prediction accuracies of Examples 1-1, 1-2, and 1-3 are almost equal. The prediction accuracies of Examples 1-4 and 1-5 are lower than those of the others. To investigate the reason for this result, Fig. 12 shows the temperature isosurface for Example 1. The sensors in Examples 1-1, 1-2, and 1-3 are arranged on three adjacent isosurfaces. The monitoring data from three sensors (sensors 3, 7, and 11) near the water inlet pipe represent the soil temperature isosurface on the water inlet side. The monitoring data from three sensors (sensors 1, 5, and 9) near the water outlet pipe represent the soil temperature isosurface on the water outlet side. The monitoring data of the remaining six sensors (sensors 2, 4, 6, 8, 10, and 12) represent the isosurface of the soil equidistant from the water inlet and outlet on both sides of the pile. While the data for these three isosurfaces are distinct, the data included inside each isosurface are continuous and close. According to the model prediction results, when the same neural network was used and the sensors were moved on the three isosurfaces, the accuracy of the energy pile prediction was almost unchanged. For Example 1-4, although the sensors were on the same isosurface, this

arrangement scheme led to too few spatial features for the network to learn and thus affected its predictive accuracy. The sensors in Example 1-5 spanned nine isosurfaces, the distance between the isosurfaces was too large, and the value changed greatly. Thus, the network cannot capture the relationship between the data, which affects its prediction performance.

**Table 9**

Performance comparison for Examples 1-1 to 1-5

|             | RMSE  | MAE   | MAPE  | $R^2$   |
|-------------|-------|-------|-------|---------|
| Example 1-1 | 0.096 | 0.078 | 0.723 | 96.252% |
| Example 1-2 | 0.088 | 0.074 | 0.666 | 96.306% |
| Example 1-3 | 0.087 | 0.073 | 0.655 | 96.192% |
| Example 1-4 | 0.154 | 0.147 | 1.351 | 88.981% |
| Example 1-5 | 0.146 | 0.129 | 1.167 | 89.560% |

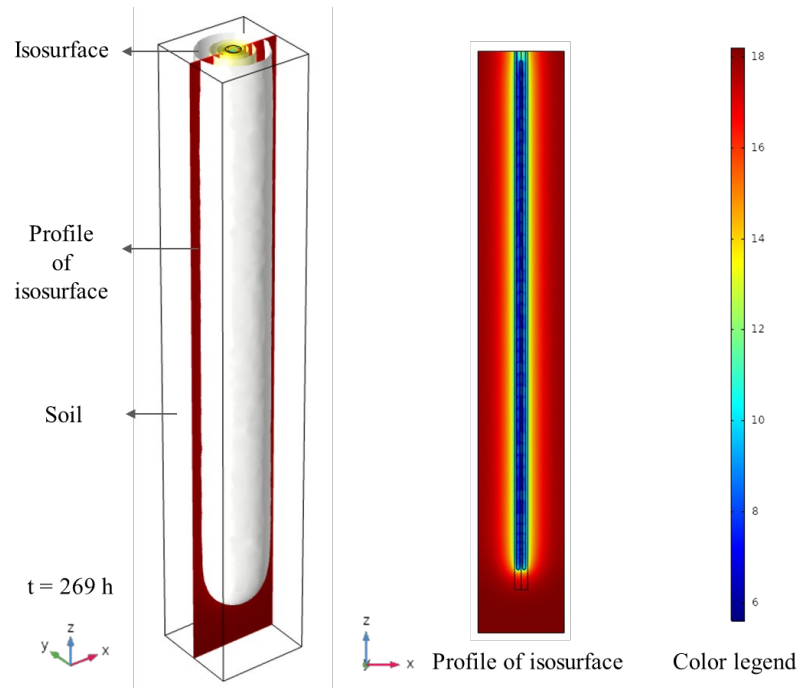


Fig. 12. Temperature isosurface of the energy pile in Example 1.

#### 4.5.2. With groundwater flow

The previous section shows that the sensor arrangement schemes shown in Examples 1-4 and 1-5 reduce the accuracy of the prediction model. Therefore, as shown in Fig. 13, to study the influence of groundwater on the STF, Examples 2-1, 2-2, and 2-3 followed the same sensor arrangement principles as those of Examples 1-1, 1-2, and 1-

3. The CNN-LSTM structures of Examples 2-1–2-3 are optimal network structures obtained through multiple tests of the model.

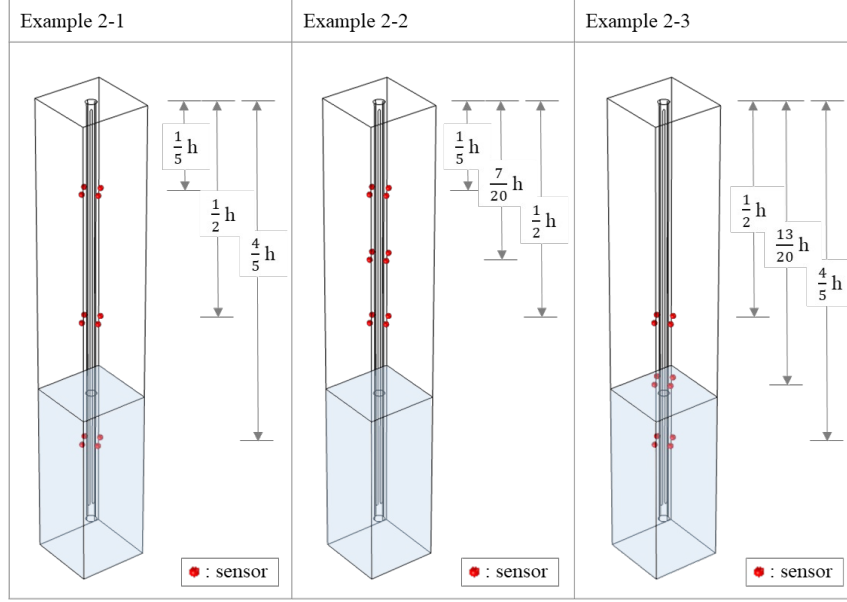


Fig. 13. Different sensor arrangement schemes in Example 2.

Table 10 shows the average values of RMSE, MAE, MAPE, and  $R^2$  for 10 consecutive training results for each network. The prediction accuracy of Examples 2-1 and 2-3 is heavily reduced compared to that of Example 2-2. The isosurface of the energy pile temperature in Example 2 (Fig. 14) is plotted. Owing to the influence of groundwater, in Examples 2-1 and 2-3, the temperature isosurface where the third group of sensors was located is too different from those where the first and second group of sensors were located, and the number of sensors was limited so that the network did not capture the features well. The data from the third group of sensors prevented the network from capturing the features well. The sensors of Example 2-2 were free from the direct influence of groundwater and were on three approximate isothermal surfaces, so CNN-LSTM learned the monitoring data well and improved the prediction accuracy. Therefore, when the sensor locations are affected by groundwater, they should be distributed on similar isothermal surfaces around the piles and should avoid being arranged in places that are directly affected by groundwater flow. Thus, the network can provide better prediction accuracy.

**Table 10**

Performance comparison for Examples 2-1-2-3

|             | RMSE  | MAE   | MAPE  | $R^2$   |
|-------------|-------|-------|-------|---------|
| Example 2-1 | 0.141 | 0.129 | 1.165 | 89.550% |
| Example 2-2 | 0.090 | 0.079 | 0.721 | 96.155% |
| Example 2-3 | 0.163 | 0.139 | 1.267 | 88.068% |

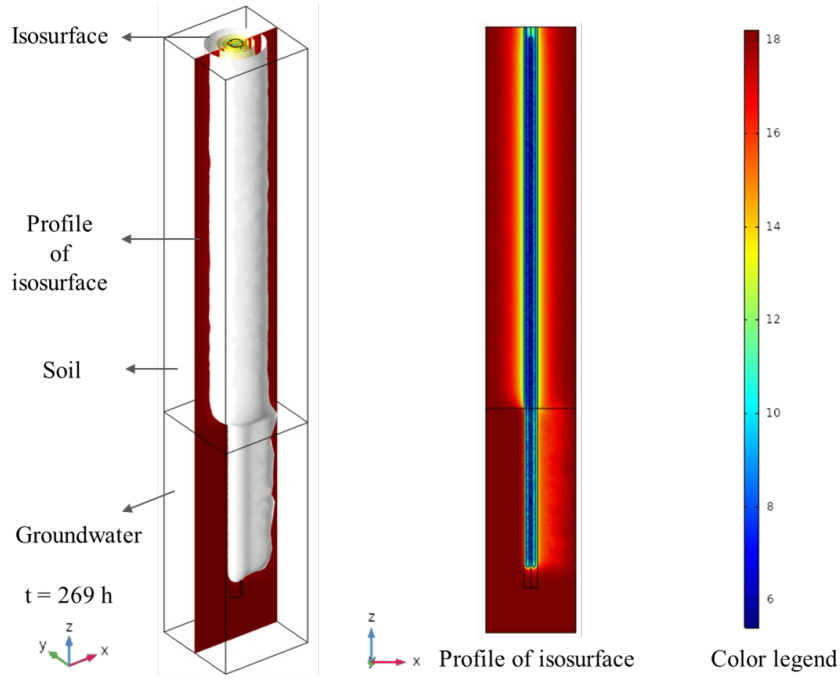


Fig. 14. Temperature isosurface of the energy pile in Example 2.

## 5. Conclusions and recommendations

Outlet water temperature prediction for energy piles can help optimize the control scheme of GHSP systems. The outlet water temperature is influenced by various dynamically changing factors, such as the constant fluctuation in the outdoor air temperature and the soil temperature around the pile. To develop an accurate and effective prediction model, it is particularly important to consider these influencing factors.

In this research, for efficient and accurate prediction of the outlet water temperature of an energy pile, a novel hybrid deep learning model CNN-LSTM was established to exploit the information in STF data. For the CNN-LSTM hybrid model, the CNN layer can extract the spatial features of the datasets, while the LSTM layer can process the time series features of the datasets. The CNN-LSTM hybrid model

exhibited a better prediction capability than the benchmark models. Specifically, when evaluating the prediction accuracy by using  $R^2$ , the CNN-LSTM model can achieve an average  $R^2$  value of 96.252% when predicting without groundwater influence. This  $R^2$  value is higher than the values of the LSTM, CNN, ANN, and SimpleRNN models by 2.326%, 3.527%, 4.585%, and 5.755%, respectively. The CNN-LSTM model can also perform effectively when the STF is influenced by groundwater flow, and an average  $R^2$  value of 96.155% can be achieved.

To further improve the prediction performance of CNN-LSTM, we build an optimal dataset. The influence mechanism of different STF monitoring sensor arrangement schemes on the network performance was specifically investigated. The results show that in the energy pile radial direction, the sensor should be arranged around the energy pile at a position 0.5 times the diameter of the pile. In the axial direction, the sensors should be arranged at a certain distance. This certain distance can be adjusted according to the actual project to ensure that all the sensors avoid being directly affected by the groundwater flow.

For future work, consider that in the practical use of energy piles, external service environments, such as the long-term accumulation of soil temperature, and internal control strategies, such as the instantaneous change in the circulating flow rate, will affect the outlet water temperature of energy piles. The influence of these factors on the prediction accuracy of the CNN-LSTM model needs to be further studied.

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# Outlet water temperature prediction of energy pile based on spatial-temporal feature extraction through CNN–LSTM hybrid model

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This study aims to develop a deep learning neural network model that can accurately predict the outlet water temperature of an energy pile according to the inlet water temperature and the soil temperature field around the energy pile. To obtain the data required for neural network training, based on the in situ energy pile experiment conducted by Gao et al. [1, 2] in Shanghai, this study established a finite element model that numerically simulated the heat transfer performance of the energy pile.

In practice, the inlet water temperature varies with the fluctuating thermal loads of a building, so a fluctuating inlet water temperature should be adopted when performing numerical simulations for an energy pile system. However, in the in situ experiments [1, 2], the inlet water temperature of the energy pile system was set to three different constant temperatures, 32 °C, 35 °C, and 38 °C. Obviously, the constant inlet water temperature cannot reflect the continuous fluctuation of the thermal load of the buildings. However, we do not have access to the in situ fluctuating inlet water temperature of the energy pile system; we used the recorded outdoor air temperature of Shanghai to represent the inlet water temperature as an alternative.

To validate the correctness of this alternative, it is essential to prove the existence of a strong correlation between the inlet water temperature of the ground heat exchanger and the outdoor air temperature. In this study, the correlation was revealed by using the Pearson correlation coefficient (PCC) method, which is a classical method used to evaluate the correlation between two sets of time series data. The principle of the PCC method together with the related calculation and analysis process are provided as follows.

### 1) Introduction of the PCC

The Pearson correlation coefficient (PCC) method is a statistical metric method used to measure the linear correlation between two variables X and Y [3]. The PCC can be calculated using Eq. (A.1) [4]. The function  $COV(X, Y)$  represents the covariance of X and Y.  $\sigma_X$  and  $\sigma_Y$  are the deviations of X and Y, respectively.  $\mu_X$  and  $\mu_Y$  are the respective means. The value of  $\rho_{X,Y}$  ranges from +1 to -1. A value of +1 indicates that X is positively linearly correlated to Y; a value of 0 indicates that X is not linearly correlated to Y; and a value of -1 implies that X is negatively linearly correlated to Y. X and Y are strongly correlated to each other when the value of  $\rho_{X,Y}$  is greater than 0.8 and are strongly correlated when  $\rho_{X,Y}$  is greater than 0.6 [4].

$$\rho_{X,Y} = \frac{COV(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y} \quad \text{Eq. (A.1)}$$

### 2) Data acquisition of inlet water temperature and outdoor air temperature:

Esen et al. [5] used ANNs to predict the performance of a GSHP system. A GSHP system with two heating ground heat exchangers (HGHEs) was built in their study, and a schematic diagram of the experimental GSHP system apparatus is shown in Fig. A.1. The inlet water temperature ( $T_{i,wa1}$ ) of the heating ground exchanger (HGHE1) and the

outdoor air temperature ( $T_a$ ) were monitored. The results are shown in Fig. A.2.

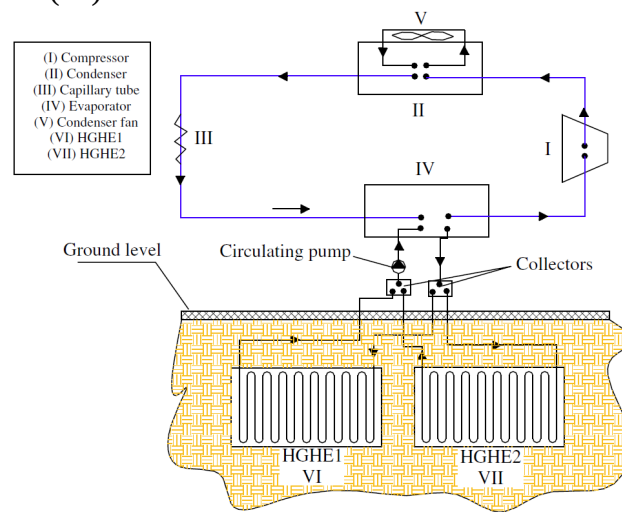


Fig. A.1. Schematic diagram of the experimental apparatus [5]

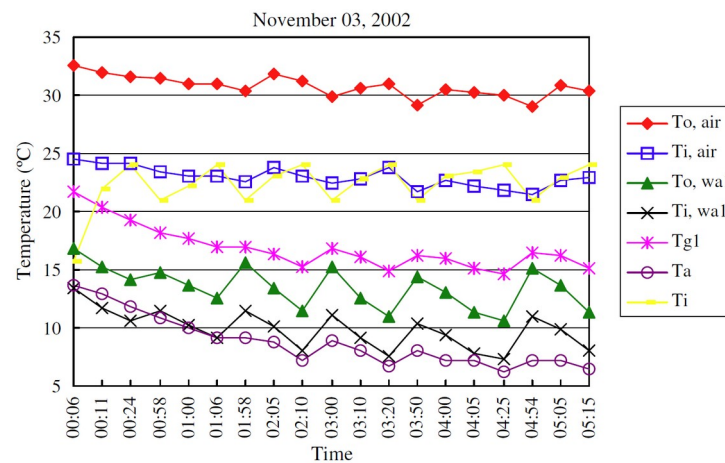


Fig. A.2. Daily variation in various temperatures for HGHE1 [5]

### 3) Calculation of the PCC

Based on the data shown in Fig. A.2, this study extracted the  $T_{i,wa1}$  and  $T_a$  data from the experiment, as shown in Table A.1:

Table A.1. Inlet water temperature and outdoor air temperature data

| Time  | $T_{i,wa1}$ (°C) | $T_a$ (°C) |
|-------|------------------|------------|
| 00:06 | 13.35            | 13.63      |
| 00:11 | 11.74            | 12.96      |
| 00:24 | 10.61            | 11.92      |
| 00:58 | 11.46            | 10.89      |
| 01:00 | 10.25            | 10.11      |

|       |       |      |
|-------|-------|------|
| 01:06 | 9.14  | 9.18 |
| 01:58 | 11.46 | 9.21 |
| 02:05 | 10.11 | 8.86 |
| 02:10 | 8.15  | 7.19 |
| 03:00 | 11.14 | 8.9  |
| 03:10 | 9.21  | 8.08 |
| 03:20 | 7.58  | 6.76 |
| 03:50 | 10.35 | 8.08 |
| 04:00 | 9.47  | 7.25 |
| 04:05 | 7.86  | 7.22 |
| 04:25 | 7.47  | 6.22 |
| 04:54 | 11    | 7.22 |
| 05:05 | 9.96  | 7.25 |
| 05:15 | 8.04  | 6.51 |

By using Python algorithms, we calculated the correlation between  $T_{i,wa1}$  and  $T_a$ , and the corresponding code is as follows:

```
import numpy as np
Ta = np.array ([13.63, 12.96, 11.92, 10.89, 10.11, 9.18, 9.21, 8.86, 7.19,
8.90, 8.08, 6.76, 8.08, 7.25, 7.22, 6.22, 7.22, 7.25, 6.51])
Tiwa1 = np.array ([13.35, 11.74, 10.61, 11.46, 10.25, 9.14, 11.46, 10.11,
8.15, 11.14, 9.21, 7.58, 10.35, 9.47, 7.86, 7.47, 11.00, 9.96, 8.04])
from scipy.stats import pearsonr
pearsonr(Ta, Tiwa1)[0]
```

By running the above code in Python, the obtained PCC for  $T_{i,wa1}$  and  $T_a$  is 0.802, which exceeds 0.8, indicating that there is an extremely strong positive linear correlation between the inlet water temperature and the outdoor air temperature in the experiment of Esen et al. [5]. In fact, when the outdoor air temperature changes, the indoor thermal load changes accordingly, which in turn causes changes in the inlet water temperature of the energy pile. Therefore, through the PCC, the inlet water temperature of the energy pile is heavily influenced by the outdoor air temperature.

#### 4) Conclusion

Through the PCC analysis, the linear correlation between the outdoor air temperature and inlet water temperature is found to be extremely strong. Therefore, it is feasible to use outdoor air temperature as an alternative for energy pile inlet water temperature in this study.

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